Results Report for

AI_DataCenter_100MW_XENDEE



POWERED BY: XENDEE

Prepared by **Jamiu Dahunsi**

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5.46%
Internal Rate of Return

XENDEE

Model Input

AI_DataCenter_100MW_CFE

Lagos, Nigeria

Objectives

Minimize cost; Ride-through outage.

Financing

Interest Rate	**0.00 %**
Investment Tax Credit	**No**
MACRS	**No**

Energy Costs

Energy Price	**$0.15 / kWh**
Avg. Natural Gas Cost	**N/A**
Avg. Diesel Fuel Cost	**$1.3 / liter**
Reference LCOE	**$0.15 / kWh**

Demand Charges

Peak TOU Rate	**N/A**
Non-Coincident	**N/A**

Demand Characteristics

Peak Demand	**20 MW**
Annual Consumption	**142.5 GWh**
Schedulable EV	**N/A**

Financial Indicators for Investment

$-66,816,900	**$-27,052,830**	**13 Years**
Project NPV (at year 10)	Project NPV (at year 20)	Payback Period

Impact

$142.9M	**-12.9%**	**76.7%**
Upfront Capital Cost	Annual Cost Reductions	Emission Savings

	Total Annual System Costs (dollars in thousands)	Total Annual CO_2 Emissions (metric tons)
Reference	$21,370.5	73,451
Investment scenario (incl. annualized capital costs and electricity sales)	$24,125.9	17,089
Total Savings (%) (incl. annualized capital costs and electricity sales)	-12.9 %	76.7 %

Result	Value
Generation-Based Levelized Cost of Electricity ($ / kWh)	*optimized*: $0.1634 / kWh *reference*: $0.1500 / kWh
Load-Served Levelized Cost of Electricity ($ / kWh)	*optimized*: $0.1693 / kWh *reference*: $0.1500 / kWh
Opex Savings (%)	69.9229%
Break-Even Year	13 years
Payback Year	13 years
IRR at End of Project (%)	5.46%
NPV at End of Project	$-27,053k

Type	Total New Capacity	Technology (New Capacity)	Init. Equip. Cost
	95.9 MW	Solar PV (95.9 MW)	88,675 k$
	183.6 MWh	Battery Energy Storage System (183.6 MWh)	54,175 k$
			Total Cost: 142,850 k$

This optimization meets the following resiliency criteria:

The supplied load will ride-through the following electric utility outage(s):

- From **May 12 12:00 AM** through **May 12 11:59 PM**

This incurs a premium cost of **$2,755,390** compared to the reference case, which is **$6.81/kWh** met or **$114,807.93 per hour**.

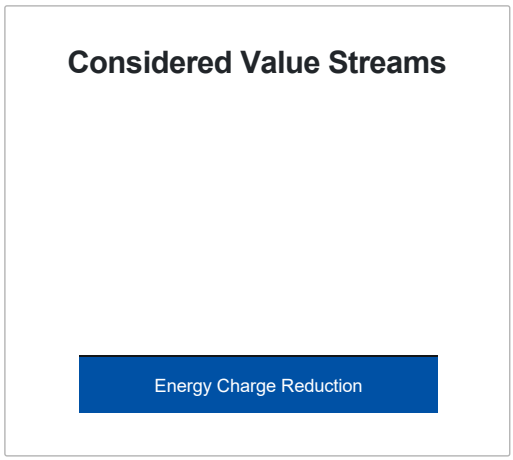
Summary

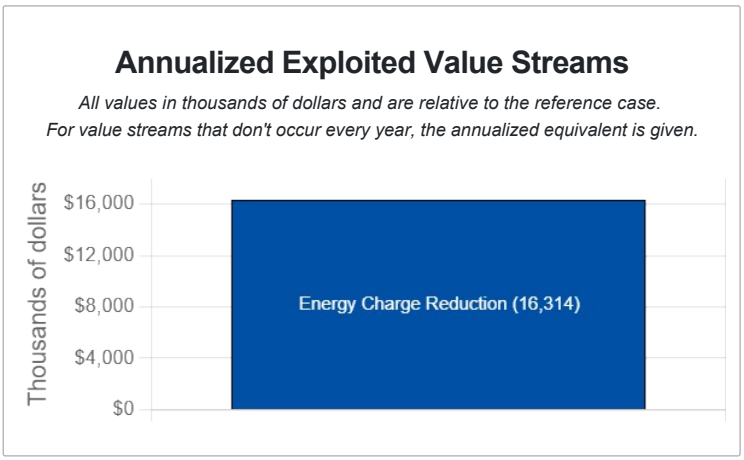
Provided in this section is an overview of projected annual costs and savings over a twenty-year period. Annualized System Costs summarizes the annualized operational and investment costs of the optimized microgrid, and the Costs and Savings Projection (Non-Discounted) presents costs as upfront investment capital, yearly operational expenses, and accumulated savings based on results from the year optimized.

Omitted Technologies

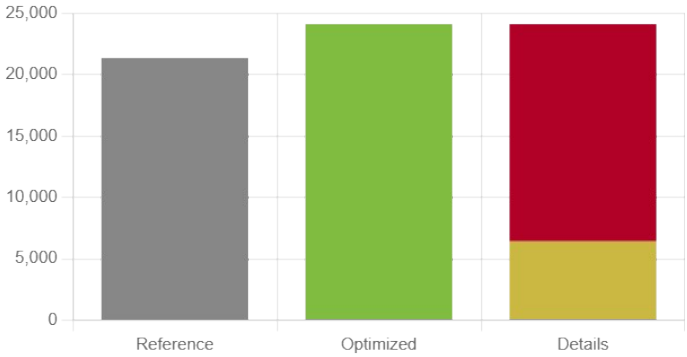
Type	Omitted Technology, Unit Size
	Diesel Generator, 2.5 MW

Value Streams



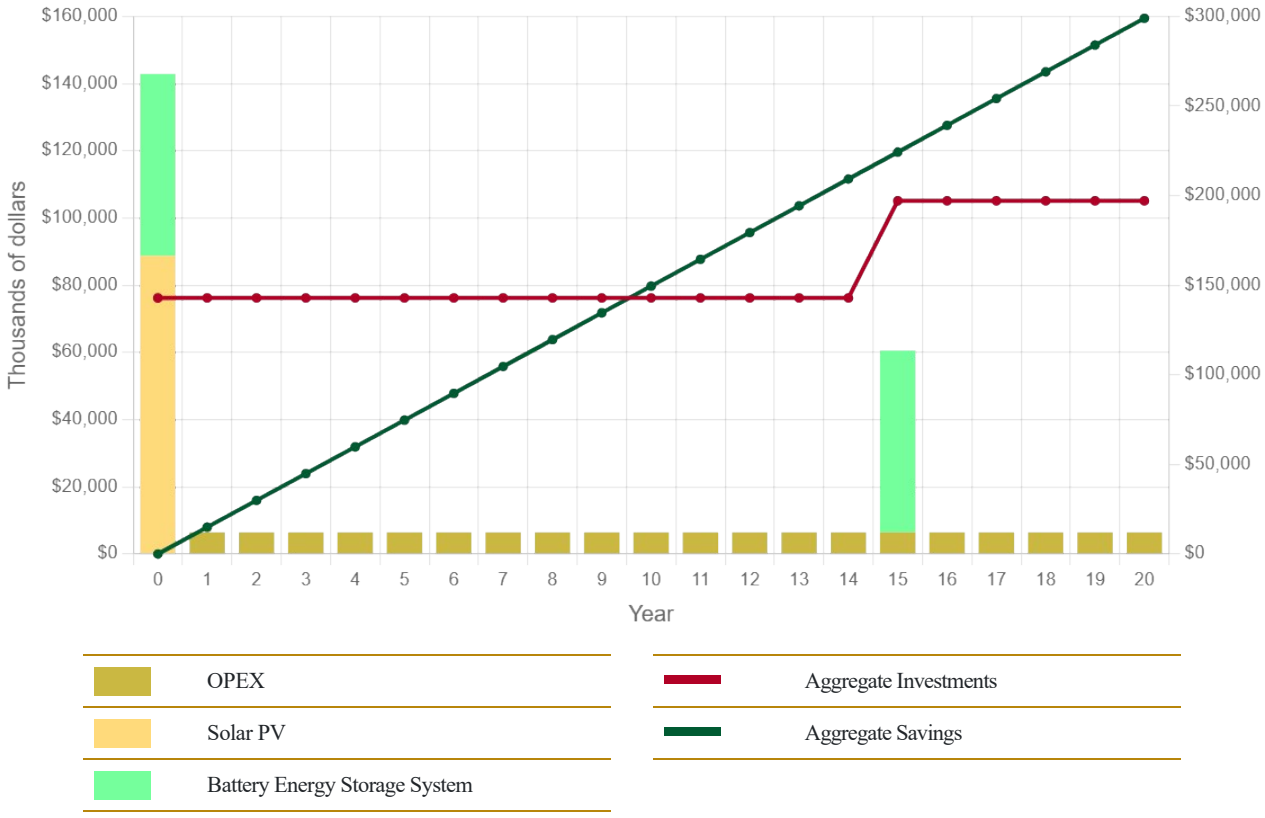


Annualized System Costs



Reference	$21,371
Optimized - Total	$24,126
OPEX	$6,428
Annualized Investment Costs	$17,698

Costs and Savings Projection (Non-Discounted)



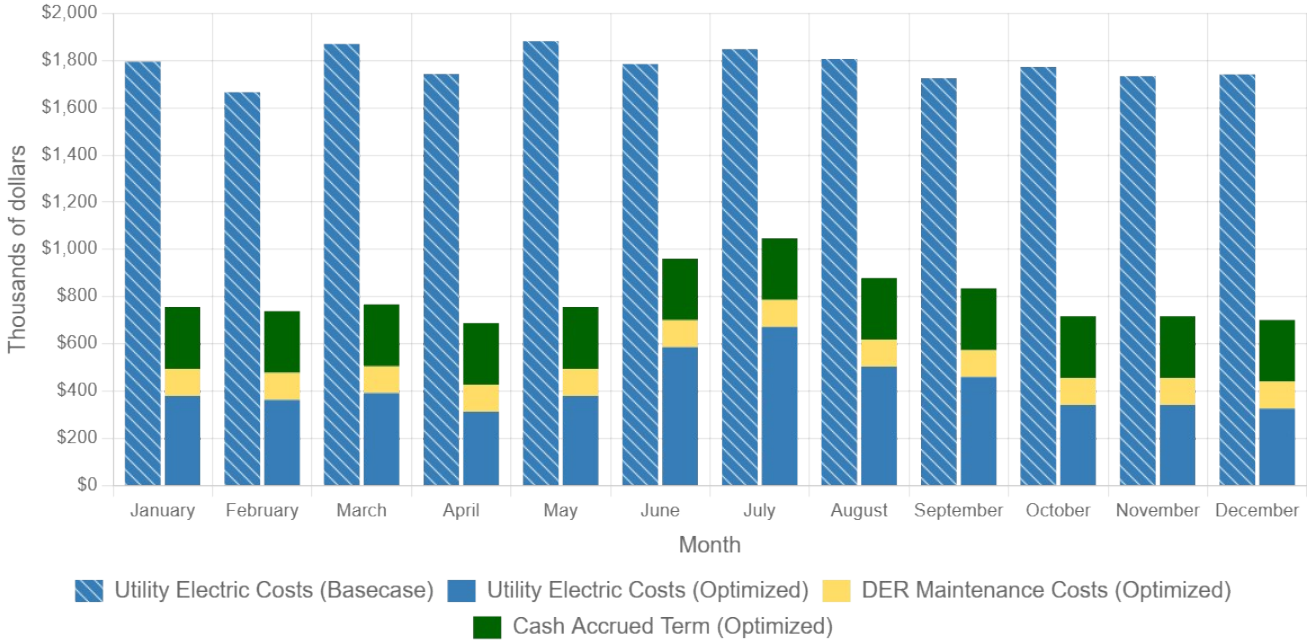
This is a non-discounted projection of the project costs and savings that assumes no changes in operation over time. Use the multi-year optimization feature to examine changes in investment and savings over time.

Note that incentives and grant funding are not included in this chart. Additionally, CAPEX are presented as the total investment costs in the year(s) corresponding to installation and/or replacement. Therefore, depending on the financing method and incentives modeled, the year when Aggregate Savings exceeds Aggregate Investments in the Costs and Savings Projection is not expected to line up with the Payback Period in the Detailed Cashflow Table.

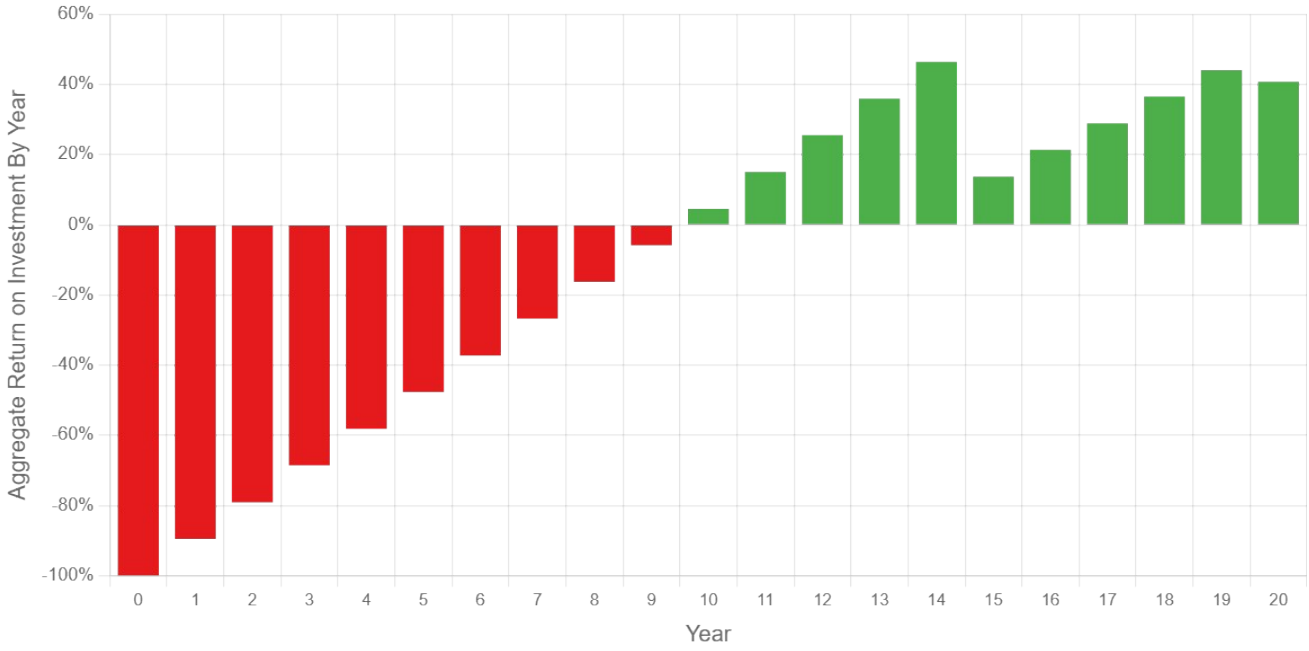
Financial Data

Primary financial indicators are provided in this section to facilitate assessing project returns. Return on investment (ROI), Net Present Value (NPV), and Internal Rate of Return (IRR) are calculated and graphed for each year leading out to twenty years from project implementation, providing insight on returns at different timelines. Also included is a detailed cash flow table.

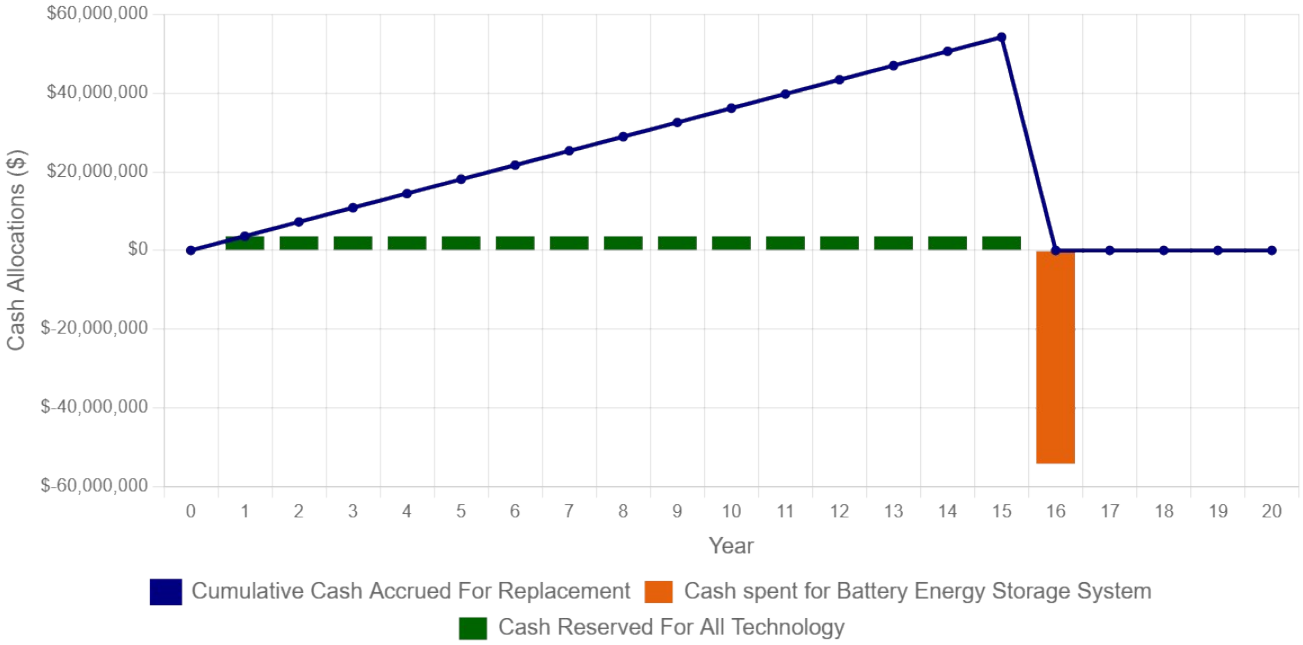
Microgrid Cost Breakdown



ROI



Cash Accrued



Detailed Cash Flow: Cost

(thousands of dollars)

The cost cashflow table below displays the **costs to run the system,** not relative to any reference. The lines shown are the costs that the solution is subject to. A positive value is a revenue while a negative is a cost. The sum of the individual cost terms is used to calculate the system **net present costs.**

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Utility Energy Charges	0.0	-5,056.9	-5,056.9	-5,056.9	-5,056.9	-5,056.9
DER Maintenance Costs	0.0	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Costs	0.0	-6,427.6	-6,427.6	-6,427.6	-6,427.6	-6,427.6
CAPEX for Solar PV	-88,674.8	0.0	0.0	0.0	0.0	0.0
CAPEX for Battery Energy Storage System	-54,175.4	0.0	0.0	0.0	0.0	0.0
Total CAPEX costs	-142,850.2	0.0	0.0	0.0	0.0	0.0
Replacement Cost for Battery Energy Storage System	0.0	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Total Replacement Costs	0.0	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Net Annual Cost (*Non-discounted*)	-142,850.2	-10,039.3	-10,039.3	-10,039.3	-10,039.3	-10,039.3
Net Annual Cost (*Discounted*)	-142,850.2	-9,295.7	-8,607.1	-7,969.5	-7,379.2	-6,832.6
Net Present Cost	**-142,850.2**	**-152,145.9**	**-160,753.0**	**-168,722.6**	**-176,101.8**	**-182,934.4**
Cumulative Cost (*Non-discounted*)	**-142,850.2**	**-152,889.6**	**-162,928.9**	**-172,968.2**	**-183,007.6**	**-193,046.9**
Cumulative Cost (*Discounted*)	**-142,850.2**	**-141,564.4**	**-139,685.3**	**-137,307.8**	**-134,516.0**	**-131,384.5**

	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11
Utility Energy Charges	-5,056.9	-5,056.9	-5,056.9	-5,056.9	-5,056.9	-5,056.9
DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Costs	-6,427.6	-6,427.6	-6,427.6	-6,427.6	-6,427.6	-6,427.6
CAPEX for Solar PV	0.0	0.0	0.0	0.0	0.0	0.0
CAPEX for Battery Energy Storage System	0.0	0.0	0.0	0.0	0.0	0.0
Total CAPEX costs	0.0	0.0	0.0	0.0	0.0	0.0
Replacement Cost for Battery Energy Storage System	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Total Replacement Costs	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Net Annual Cost (*Non-discounted*)	-10,039.3	-10,039.3	-10,039.3	-10,039.3	-10,039.3	-10,039.3
Net Annual Cost (*Discounted*)	-6,326.5	-5,857.9	-5,423.9	-5,022.2	-4,650.2	-4,305.7
Net Present Cost	**-189,260.8**	**-195,118.7**	**-200,542.6**	**-205,564.8**	**-210,214.9**	**-214,520.6**
Cumulative Cost (*Non-discounted*)	**-203,086.2**	**-213,125.5**	**-223,164.9**	**-233,204.2**	**-243,243.5**	**-253,282.8**
Cumulative Cost (*Discounted*)	**-127,978.8**	**-124,356.7**	**-120,569.0**	**-116,660.2**	**-112,668.8**	**-108,628.7**

	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17
Utility Energy Charges	-5,056.9	-5,056.9	-5,056.9	-5,056.9	-5,056.9	-5,056.9
DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Costs	-6,427.6	-6,427.6	-6,427.6	-6,427.6	-6,427.6	-6,427.6
CAPEX for Solar PV	0.0	0.0	0.0	0.0	0.0	0.0
CAPEX for Battery Energy Storage System	0.0	0.0	0.0	0.0	0.0	0.0
Total CAPEX costs	0.0	0.0	0.0	0.0	0.0	0.0
Replacement Cost for Battery Energy Storage System	-3,611.7	-3,611.7	-3,611.7	-3,611.7	0.0	0.0
Total Replacement Costs	-3,611.7	-3,611.7	-3,611.7	-3,611.7	0.0	0.0
Net Annual Cost (*Non-discounted*)	-10,039.3	-10,039.3	-10,039.3	-10,039.3	-6,427.6	-6,427.6
Net Annual Cost (*Discounted*)	-3,986.8	-3,691.4	-3,418.0	-3,164.8	-1,876.2	-1,737.2
Net Present Cost	**-218,507.4**	**-222,198.8**	**-225,616.8**	**-228,781.7**	**-230,657.8**	**-232,395.0**
Cumulative Cost (*Non-discounted*)	**-263,322.2**	**-273,361.5**	**-283,400.8**	**-293,440.2**	**-299,867.8**	**-306,295.4**
Cumulative Cost (*Discounted*)	**-104,568.9**	**-100,514.5**	**-96,486.9**	**-92,504.6**	**-87,528.6**	**-82,782.1**

	Year 18	Year 19	Year 20
Utility Energy Charges	-5,056.9	-5,056.9	-5,056.9
DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8
Total OPEX Costs	-6,427.6	-6,427.6	-6,427.6
CAPEX for Solar PV	0.0	0.0	0.0
CAPEX for Battery Energy Storage System	0.0	0.0	0.0
Total CAPEX costs	0.0	0.0	0.0
Replacement Cost for Battery Energy Storage System	0.0	0.0	0.0
Total Replacement Costs	0.0	0.0	0.0
Net Annual Cost (*Non-discounted*)	-6,427.6	-6,427.6	-6,427.6
Net Annual Cost (*Discounted*)	-1,608.5	-1,489.4	-1,379.0
Net Present Cost	**-234,003.5**	**-235,492.9**	**-236,871.9**
Cumulative Cost (*Non-discounted*)	**-312,723.1**	**-319,150.7**	**-325,578.3**
Cumulative Cost (*Discounted*)	**-78,258.6**	**-73,951.1**	**-69,852.3**

Detailed Cash Flow: Savings

(thousands of dollars)

The Savings cashflow table below displays the **savings the system produces relative to the reference.** The lines shown are the savings that the solution creates. A positive value is a savings while a negative is a loss. The sum of the individual savings terms is used to calculate the **net present value** of the system.

	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Savings: Utility Energy Charges	0.0	16,313.7	16,313.7	16,313.7	16,313.7	16,313.7
Savings: DER Maintenance Costs	0.0	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Savings	0.0	14,942.9	14,942.9	14,942.9	14,942.9	14,942.9
CAPEX difference for Solar PV	-88,674.8	0.0	0.0	0.0	0.0	0.0
CAPEX difference for Battery Energy Storage System	-54,175.4	0.0	0.0	0.0	0.0	0.0
Total CAPEX Difference	-142,850.2	0.0	0.0	0.0	0.0	0.0
Replacement Cost Difference for Battery Energy Storage System	0.0	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Total Replacement Costs Difference	0.0	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Net Annual Cash Flow (*Non-discounted*)	-142,850.2	11,331.2	11,331.2	11,331.2	11,331.2	11,331.2
Net Annual Cash Flow (*Discounted*)	-142,850.2	10,491.9	9,714.7	8,995.1	8,328.8	7,711.8
Net Present Value	**-142,850.2**	**-132,358.4**	**-122,643.7**	**-113,648.6**	**-105,319.8**	**-97,608.0**
Cumulative Cash Flow (*Non-discounted*)	**-142,850.2**	**-131,519.0**	**-120,187.8**	**-108,856.6**	**-97,525.4**	**-86,194.2**
Cumulative Cash Flow (*Discounted*)	**-142,850.2**	**-121,776.9**	**-103,041.7**	**-86,413.9**	**-71,684.1**	**-58,662.3**

	Year 6	Year 7	Year 8	Year 9	Year 10	Year 11
Savings: Utility Energy Charges	16,313.7	16,313.7	16,313.7	16,313.7	16,313.7	16,313.7
Savings: DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Savings	14,942.9	14,942.9	14,942.9	14,942.9	14,942.9	14,942.9
CAPEX difference for Solar PV	0.0	0.0	0.0	0.0	0.0	0.0
CAPEX difference for Battery Energy Storage System	0.0	0.0	0.0	0.0	0.0	0.0
Total CAPEX Difference	0.0	0.0	0.0	0.0	0.0	0.0
Replacement Cost Difference for Battery Energy Storage System	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Total Replacement Costs Difference	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7	-3,611.7
Net Annual Cash Flow (*Non-discounted*)	11,331.2	11,331.2	11,331.2	11,331.2	11,331.2	11,331.2
Net Annual Cash Flow (*Discounted*)	7,140.6	6,611.7	6,121.9	5,668.4	5,248.5	4,859.8
Net Present Value	**-90,467.4**	**-83,855.8**	**-77,733.9**	**-72,065.4**	**-66,816.9**	**-61,957.1**
Cumulative Cash Flow (*Non-discounted*)	**-74,863.0**	**-63,531.8**	**-52,200.6**	**-40,869.4**	**-29,538.1**	**-18,206.9**
Cumulative Cash Flow (*Discounted*)	**-47,176.4**	**-37,070.2**	**-28,202.3**	**-20,444.9**	**-13,681.9**	**-7,808.6**

	Year 12	Year 13	Year 14	Year 15	Year 16	Year 17
Savings: Utility Energy Charges	16,313.7	16,313.7	16,313.7	16,313.7	16,313.7	16,313.7
Savings: DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8	-1,370.8
Total OPEX Savings	14,942.9	14,942.9	14,942.9	14,942.9	14,942.9	14,942.9
CAPEX difference for Solar PV	0.0	0.0	0.0	0.0	0.0	0.0
CAPEX difference for Battery Energy Storage System	0.0	0.0	0.0	0.0	0.0	0.0
Total CAPEX Difference	0.0	0.0	0.0	0.0	0.0	0.0
Replacement Cost Difference for Battery Energy Storage System	-3,611.7	-3,611.7	-3,611.7	-3,611.7	0.0	0.0
Total Replacement Costs Difference	-3,611.7	-3,611.7	-3,611.7	-3,611.7	0.0	0.0
Net Annual Cash Flow (*Non-discounted*)	11,331.2	11,331.2	11,331.2	11,331.2	14,942.9	14,942.9
Net Annual Cash Flow (*Discounted*)	4,499.8	4,166.5	3,857.8	3,572.1	4,361.7	4,038.6
Net Present Value	**-57,457.4**	**-53,290.9**	**-49,433.1**	**-45,861.0**	**-41,499.3**	**-37,460.7**
Cumulative Cash Flow (*Non-discounted*)	**-6,875.7**	**4,455.5**	**15,786.7**	**27,117.9**	**42,060.8**	**57,003.7**
Cumulative Cash Flow (*Discounted*)	**-2,730.5**	**1,638.3**	**5,374.8**	**8,548.7**	**12,277.2**	**15,406.3**

	Year 18	Year 19	Year 20
Savings: Utility Energy Charges	16,313.7	16,313.7	16,313.7
Savings: DER Maintenance Costs	-1,370.8	-1,370.8	-1,370.8
Total OPEX Savings	14,942.9	14,942.9	14,942.9
CAPEX difference for Solar PV	0.0	0.0	0.0
CAPEX difference for Battery Energy Storage System	0.0	0.0	0.0
Total CAPEX Difference	0.0	0.0	0.0
Replacement Cost Difference for Battery Energy Storage System	0.0	0.0	0.0
Total Replacement Costs Difference	0.0	0.0	0.0
Net Annual Cash Flow (*Non-discounted*)	14,942.9	14,942.9	14,942.9
Net Annual Cash Flow (*Discounted*)	3,739.5	3,462.5	3,206.0
Net Present Value	**-33,721.3**	**-30,258.8**	**-27,052.8**
Cumulative Cash Flow (*Non-discounted*)	**71,946.6**	**86,889.5**	**101,832.4**
Cumulative Cash Flow (*Discounted*)	**18,004.6**	**20,133.4**	**21,848.0**

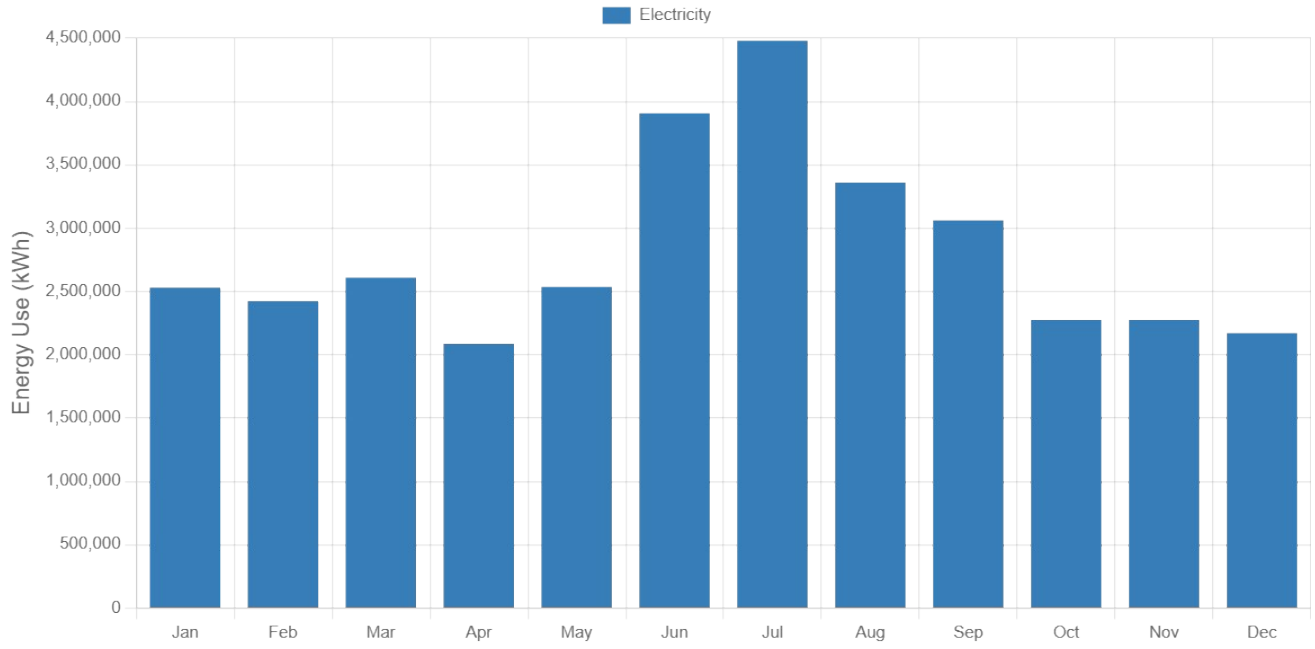
Utility Data

This section provides a summary of electricity and fuel utility purchases. Monthly breakdowns of energy consumption [kWh], demand by time-of-use period [kW], and total charges [k$] are included.

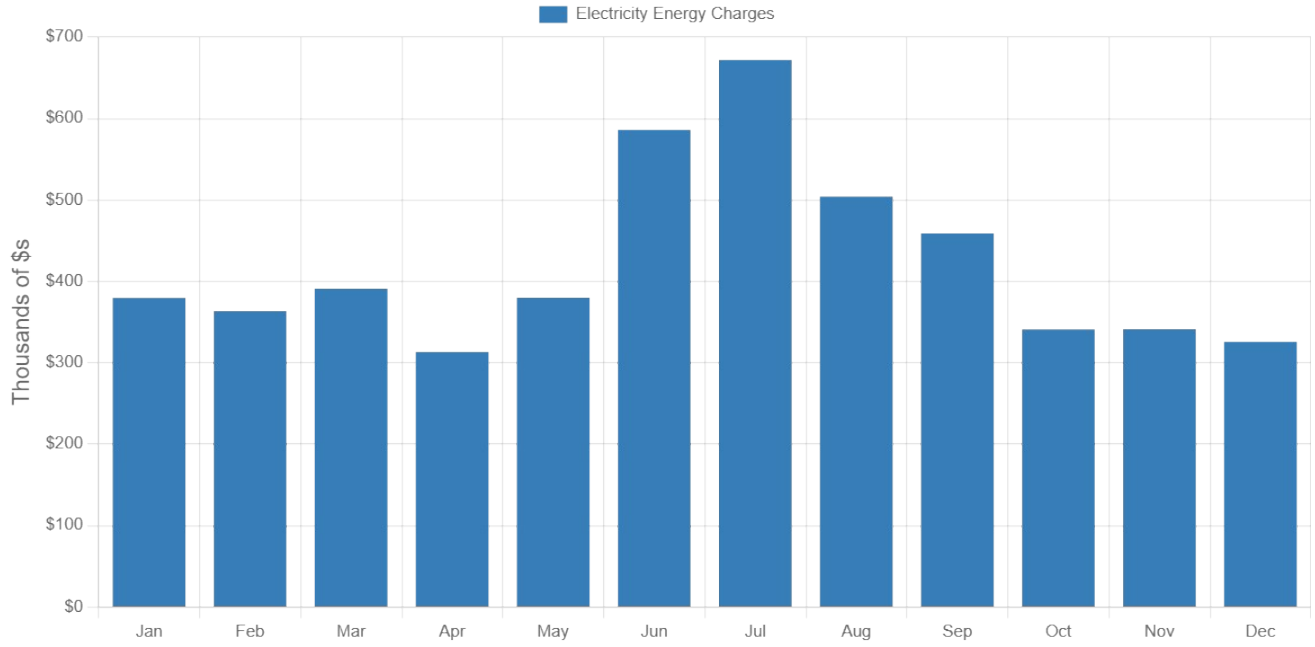
Monthly Demand (kW)



Monthly Energy Consumption (kWh)



Monthly Utility Charge Breakdown



Utility Billing Period

Billing for Annual

Annual Summary of Charges

Electricity Energy Charges [$]	5,056,884
Electricity Monthly Fee [$]	0
Fuel Purchase Costs [$]	0
Total [$]	5,056,884
Reference [$]	21,370,537
Savings [$]	16,313,653

Annual Fuel Charges

Fuel Category	Consumption [kWh]	Rate [$/kWh]	Fuel Charge [$]
Fuel Subtotal [$]			0
Reference [$]			0
Savings [$]			0

Annual Electricity Charges

Tariff	Energy Category	Consumption [kWh]	Rate [$/kWh]	Energy Charge [$]
NERC_Band_A_Lagos_Tariff	Period 1	33,712,561	-	5,056,884
Energy Subtotal [$]				5,056,884
Reference [$]				21,370,537
Savings [$]				16,313,653

Demand Category	Demand [kW]	Rate [$/kW]	Demand Charge [$]
Demand Subtotal [$]			0
Reference [$]			0
Savings [$]			0

Energy Balance and Technology Investments

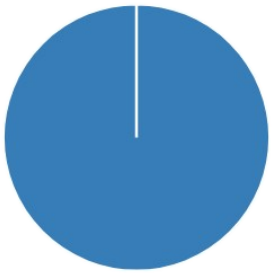
This section provides data on system energy demand and portfolio technologies. Included are details on total annual demand for each end-use modeled, share of demand met by utility purchases and on-site DER assets, total capacities of existing and new DER assets, and upfront investment costs.

Annual Electricity Balance (kWh)



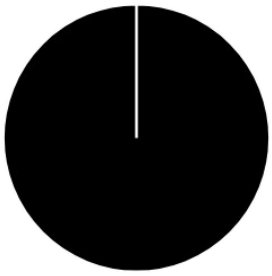
Total annual electricity purchase (kWh)	33,712,561
Total annual on-site generation from renewables (kWh)	113,981,464
Total	**147,694,025**

Utility Balance (kWh)



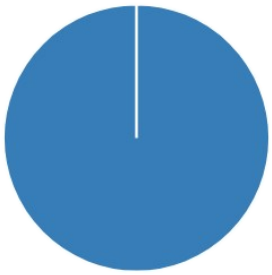
Total annual electricity purchase (kWh)	33,712,561

Aggregated Demand (kWh)



Electricity-Only Demand	142,476,830

CO_2 Emissions (metric tons)

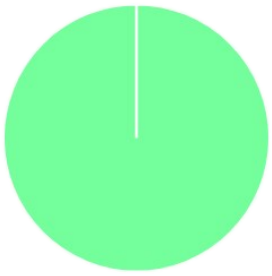


Electricity	17,089
Total	**17,089**

Generation Technologies (kWh)

New Solar PV (kWdc)	95,865
Total	**95,865**

Storage Technologies



New Battery Energy Storage System Capacity (kWh)	183,645
Total	**183,645**

Tech	Inverter Size (kW): Calc. / Specified	Ratio
Battery Energy Storage System	39,150 / 45,911	0.85

Investments

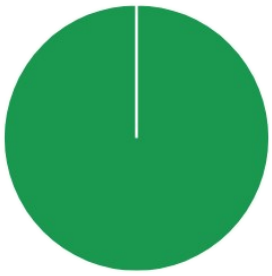


New Solar PV	$88,674,832
New Battery Energy Storage System	$54,175,401
Total	**$142,850,233**

Annual PV Electricity Balance (kWh)

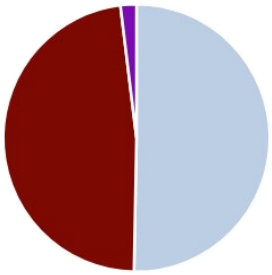
Electricity Consumed On-site	113,981,464
Electricity Curtailed	11,259,608
Total	**125,241,071**

CAPEX Breakdown ($)



Total Paid Upfront	$142,850,233
Total	**$142,850,233**

Electric Technology Losses (kWh)



Electric Battery Charging	2,665,113
Electric Battery Discharging	2,530,793
Electric Battery Standby	102,339

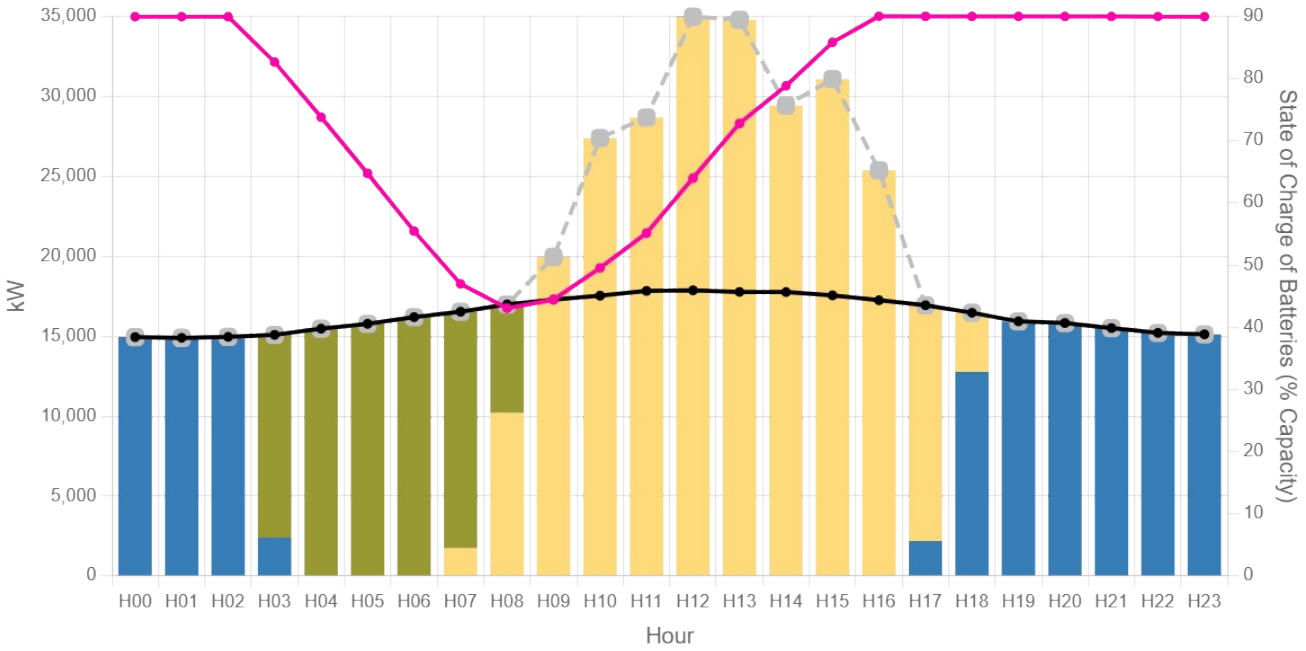
Space Used

Tech	Used / Available Space	%
Solar PV	445,882 / 929,023 m^2	48.00%

Electricity Dispatch

The following dispatch curves show the optimal system operation to meet all electricity loads on a selection of modeled days. Electricity dispatch shows both the electricity-only loads and any electricity used to operate cooling and/or refrigeration technologies. System operation includes on-site generation and storage dispatch, utility purchases, and load management strategies.

Electricity Dispatch for July, Week



Demand lines are stacked to show the cumulative impact.

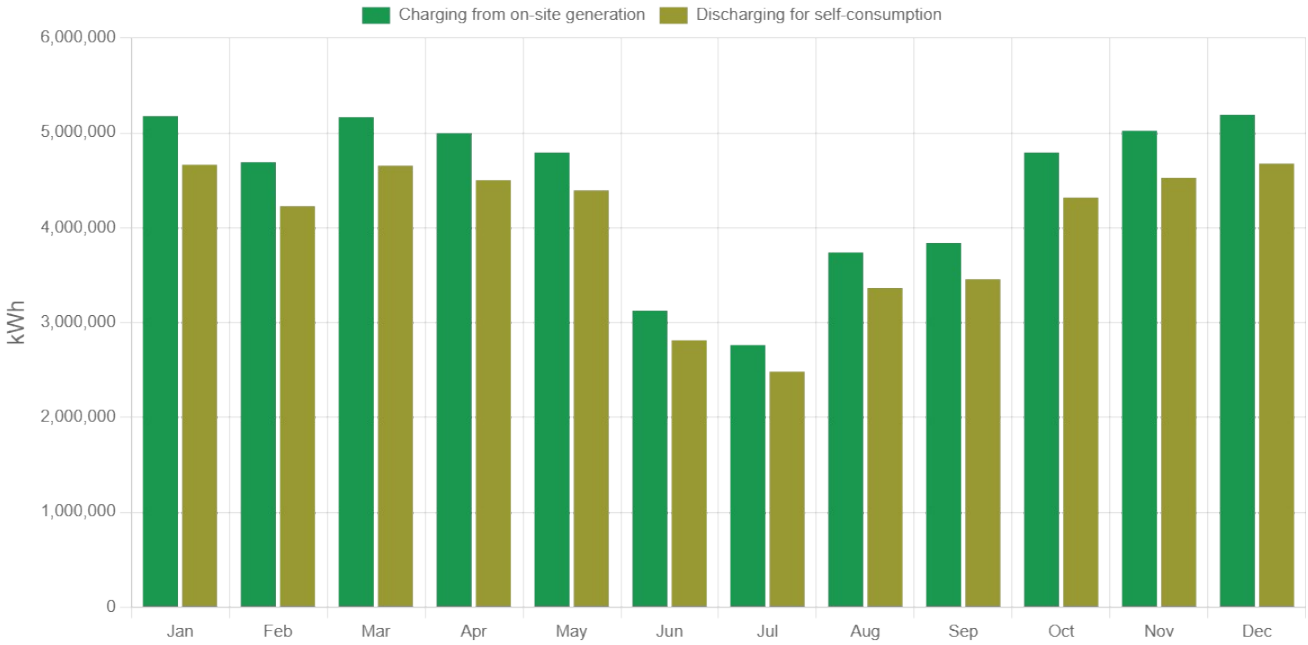
- Electricity-Only Load
- Electricity Supplied to Storage
- Utility Purchase
- Solar PV for Self Consumption
- Battery Energy Storage System for Self Consumption
- Solar PV Curtailment
- State of Charge of Battery Energy Storage System

* Axes NOT Scaled on Dispatch Graph By Data Across All Months / Day Types

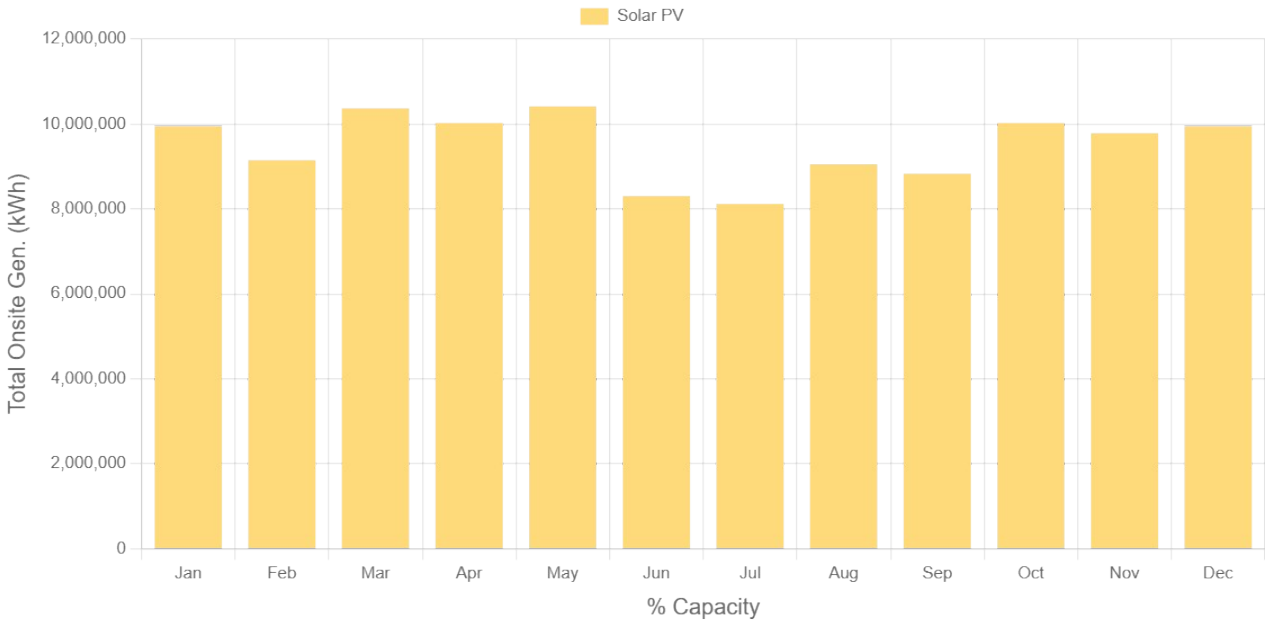
Operation Summary

This section provides a summary of generator operation and monthly on-site generation.

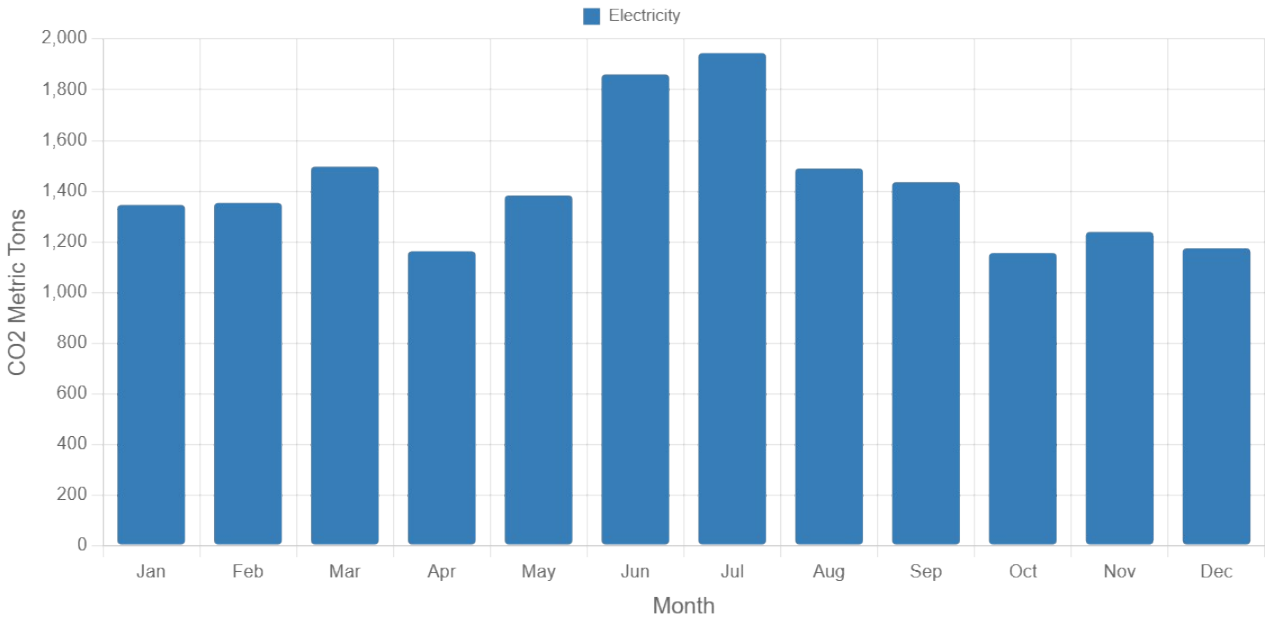
Battery Operations Summary



Monthly On-Site Generation (kWh)



Monthly Emissions Summary



Technology Capacity Factor

Tech	Capacity Factor
Solar PV	13.57%

Storage Cycling

Tech	Annual / Allowed Cycles	%
Battery Energy Storage System	276 / N/A	N/A